

Brandon M. Booth

Assistant Professor of Computer Science @ University of Memphis
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Last Updated: March 13, 2026

Education

PhD in Computer Science <i>University of Southern California</i> Advisor: Shrikanth S. Narayanan Title: <i>Improving Modeling of Human Experience and Behavior: Methodologies for Enhancing the Quality of Human-produced Data and Annotations of Subjective Constructs</i>	2014–2020
M.S in Computer Science <i>University of Southern California</i> Emphasis on Game Development	2008–2011
B.S. Computer Science and B.S. in Applied Mathematics <i>University of Colorado Boulder</i>	2003–2007

Professional Appointments

Academic

Assistant Professor of Computer Science <i>Department of Computer Science (full time)</i> University of Memphis	2023–present
Research Scientist II <i>Institute of Cognitive Science (full time)</i> University of Colorado Boulder	2022–2023
Research Scientist <i>Institute of Cognitive Science (full time)</i> University of Colorado Boulder	2020–2022
Research Assistant <i>Signal Analysis and Interpretation Laboratory (part time)</i> University of Southern California	2015–2020

Industry and Research

Senior Software Engineer <i>Applied Minds (full time)</i> Burbank, CA, USA	2012–2015
Graphics and Core Software Engineer <i>Seed Studio Inc. (full time)</i> Los Angeles, CA, USA	2012 Spring
Game and Engine Software Engineer <i>Heavy Iron Studios (full time)</i> Los Angeles, CA, USA	2008–2011
Programming Intern <i>Wideload Games Inc. (full time)</i> Chicago, IL, USA	2007 Summer

Software Engineer

2004-2005

Applied Math Department (part time)
University of Colorado Boulder

Teaching

Assistant Professor of Computer Science

2023-present

University of Memphis (full time)
Memphis, TN, USA

Guest Lecturer and Teaching Assistant

2014-2015

University of Southern California (part time)
Los Angeles, CA, USA

Game Programming Instructor

2011 Summer/Fall

Art Institute of Los Angeles (part time)
Los Angeles, CA, USA

Awards and Honors

2025: UOFM DIVISION OF RESEARCH AND INNOVATION: First-time Principal Investigator Award

2023: ACM ICMI CONFERENCE: Outstanding Reviewer Award

2023: IEEE ICASSP CONFERENCE: Outstanding Reviewer Award

2021: ACM ICMI CONFERENCE: Top 5% Best Reviewer Award

2021: ACM UMAP CONFERENCE: Best Reviewer Award

2019: USC STEVENS CENTER FOR INNOVATION: Commercialization Award (TILES Audio Recorder)

2018: AUDIO/VISUAL EMOTION CHALLENGE (AVEC) WORKSHOP: Gold-standard Emotion Sub-challenge Winner

External Funding

In Preparation/Under Review

STAR-AI: Scalable Training for Responsible Artificial Intelligence

2026

Agency: NSF CyberTraining

PI: Booth (PI)

Amount: \$749,997 requested

Status: Under review

Modeling Motivational Trajectories in Digital Learning Systems

2026

Agency: National Science Foundation (CISE Core / ECR)

PI: Booth (PI)

Amount: \$599,894 requested

Status: Under review

Funded

Multi-Agent AI for Critical Thinking in Middle School Project-Based Learning

2026

Agency: UofM Community of AI Researchers (CoAIRS)

PI: Phan (PI), Booth (Co-PI), Cook (Co-PI)

Amount: \$20k awarded

Period: 2026

Inferring Motivational Profiles from Digital Learning Behaviors in an Intelligent Tutoring System

2025

Agency: UofM Institute for Intelligent Systems' Affiliate Summer Research Fellowship

PI: Booth (PI), Harrell-Williams, L., Windsor, A., Pavlik, P., Kuhlmann, S.

Amount: \$5k awarded

Period: Summer 2025

Generative Artificial Intelligence Facilitated After-Action Reviews for Military Training Agency: Office of Naval Research PI: Hickman (PI Virginia Tech), Booth (Co-PI), McMahan (Co-PI Virginia Tech) Amount: \$794,985 awarded (Co-PI Booth subaward \$262,803) Period: 2025-2027	2025
NSF Student Travel Grant for 2024 ACM International Conference on Multimodal Interaction (ACM ICMI) Agency: National Science Foundation (IIS #2430047) PI: Booth (PI) Amount: \$12.2k awarded Period: 2024-2025	2024
Emotional State and Calibrated Trust in Human-Agent Teams: A Systematic Literature Review and Accompanying Meta-Analysis Agency: Toyota Research Institute PI: Hirshfield (PI University of Colorado Boulder), D'Mello (Co-PI UCB), Booth (Co-PI) Amount: \$54k awarded Period: 2022-2023	2022
Hybrid Human-Agent Tutoring (HAT) Platform to Accelerate Middle School Math Achievement for Low Income Students Agency: Various non-profits (withheld at request of funders) PI: D'Mello (PI University of Colorado Boulder), Sumner (Co-PI UCB), Booth (Senior Personnel) Amount: \$8.9M awarded Period: 2022-2027 Not Awarded	2022
A Clinical Trial of Virtual Reality Using Cue Exposure: An Innovative Technology for Tobacco Cessation Among African American Dual Users Agency: National Institutes of Health (NIH) PI: El Hajj (Lead PI), Booth (Co-I), Brodt (Co-I), Jiang (Co-I), Haddad (Co-I) Amount: \$385,834 (requested)	2025
Collaborative Research: HCC: Small: Enhancing Human-AI Collaboration in High-stakes Selection: Integrating Human Preferences with Ratings in Video Assessments Agency: National Science Foundation (CISE) PI: Booth (PI), Hickman (Co-PI Virginia Tech), D'Mello (Co-PI University of Colorado Boulder) Amount: \$326,769 (requested)	2025
Exploring In-the-Moment Motivation Profiles and Learning Behaviors During Mathematics Learning in MATHia Agency: National Science Foundation PI: Williams (PI University of Memphis), Windsor (Co-PI), Pavlik (Co-PI), Booth (Co-PI), Kuhlmann (Co-PI) Amount: \$1,498,693 (requested)	2025
Longitudinally Predicting and Tracing Self-Regulated Learning Across STEM Gateway Courses to Enhance Persistence and Success in STEM Majors Agency: National Science Foundation (ECR Core) PI: Kuhlmann (PI University of Memphis), DeYonker (Co-PI), Booth (Co-PI), Abell (Co-PI), Sabatini (Co-PI), Johnson (Co-PI) Amount: \$1,260,981 (requested)	2025

Planning Grant: Tracing Middle School Students Self-Regulated Learning in Autonomous Video-Based Learning Environments with Digital Data from Khan Academy	2025
<i>Agency:</i> Bill and Melinda Gates Foundation (BMGF)	
<i>PI:</i> Kuhlmann (PI University of Memphis), Booth (Co-PI)	
<i>Amount:</i> \$10k (requested)	
US Spanish: A Gamified Introduction Through an App	2025
<i>Agency:</i> National Endowment for the Humanities (NEH)	
<i>PI:</i> Ortega-Santos (PI University of Memphis), Toledo (Co-PI), Vann (Co-PI), Booth (Co-PI)	
<i>Amount:</i> \$27.6k (requested)	
Developing Self-regulated, Engaged, Adaptive Learners: Transforming Post-secondary Instruction Through GenAI	2025
<i>Agency:</i> Institute of Educational Sciences (IES)	
<i>PI:</i> Sabatini (PI University of Memphis), Rus (Co-PI), Pavlik (Co-PI), Kuhlmann (Co-PI), Booth (Co-PI)	
<i>Amount:</i> \$1,698,859 (requested)	
CyberTraining: Implementation: Small: Empowering Trustworthy AI for Multidisciplinary Sciences on GPU Cyberinfrastructure	2025
<i>Agency:</i> National Science Foundation (CyberTraining)	
<i>PI:</i> Huang (PI University of Memphis), Booth (Co-PI)	
<i>Amount:</i> \$500k (requested)	
CogniVideo: An AI-enabled Adaptive Web-based Video Player for Personalized Self-regulated Learning	2024
<i>Agency:</i> The Learning Academy (Tools Competition)	
<i>PI:</i> Booth (PI), Kuhlmann (PI University of Memphis)	
<i>Amount:</i> \$50k (requested)	
Exploring In-the-Moment Motivation Profiles and Learning Behaviors During Mathematics Learning in MATHia	2024
<i>Agency:</i> Institute of Educational Sciences	
<i>PI:</i> Williams (PI University of Memphis), Windsor (Co-PI), Pavlik (Co-PI), Booth (Co-PI), Kuhlmann (Co-PI)	
<i>Amount:</i> \$688.6k (requested)	
Modeling High-stakes Selection using Forced Ordinal Preferences to Uncover Implicit and Algorithmic Biases: The Case of Automated Video Interviews	2023
<i>Agency:</i> National Science Foundation (NSF 23-561)	
<i>PI:</i> Booth (PI), D'Mello (Co-PI University of Colorado Boulder), Hickman (Co-PI Virginia Tech)	
<i>Amount:</i> \$591.7k (requested)	
Responsive Approaches In Science Exhibits (RAISE): Optimizing Productive Struggle for Empowered Learning	2021
<i>Agency:</i> National Science Foundation (NSF 20-607)	
<i>PI:</i> Kipling (PI Boston Museum of Science), D'Mello (Co-PI University of Colorado Boulder), May (Co-PI Boston Museum of Science), Rappolt-Schlichtmann (Co-PI EdTogether), Booth (Senior Personnel)	
<i>Amount:</i> \$2M (requested)	
<i>Review:</i> Highly Competitive	

Publications

Acceptance rates (AR), five-year impact factors (IF), or H5 indices (H5) reported where available.

Journals

2026: Kuhlmann S., **Booth B.M.**, Patterson C., Siegfried H., Hollander-Blackmon C., Yenumula S.S., Bernacki M.L.
 "Using high school students' prior knowledge to personalize formative assessments within instructional videos that teach

the science of learning." Learning and Individual Differences, Special Issue on Adaptive Formative Assessments. (abstract accepted; manuscript submitted)

2026: **Booth B.**, Jung Y., Tran K., Dadhich S., Phan V. "Polylogue: A Multi-Agent Framework for Supporting Critical Thinking Through Empirically Grounded Conversations." International Journal of Educational Technology in Higher Education, 2025. (submitted; H5=54)

2025: Feng T., **Booth B.**, Mundnich K., Zhou E., Girault B., Lerman K., Narayanan S. "TILES-2018 Sleep Eval Dataset: A Longitudinal Wearable Sleep Dataset of Hospital Workers for Understanding Sleep Patterns." Journal of Behavior and Health Informatics, 2025. (submitted; IF=3.1)

2024: Kerschbaumer S., Voracek M., Aczél B., Anderson S.F., **Booth B.M.**, Buchanan E.M., Carlsson R., Heck D.W., Pauliina A., Hiekkaranta R.H., Karch J.D. "VALID: a checklist-based approach for improving validity in psychological research." Adv. Meth. Pract. Psychol. Sci. 2024. (H=42)

2024: Hickman L., Huynh C., Gass J., **Booth B.**, Kuruzovich J., Tay L. "Whither Bias Goals, I Will Go: An Integrative Systematic Review of Algorithmic Bias Mitigation." Journal of Applied Psychology, 2024. (H=340)

2024: **Booth B.**, Narayanan S. "People Make Mistakes: Obtaining Accurate Ground Truth from Continuous Annotations of Subjective Constructs." Behavior Research Methods, 2024. (IF=7.2)

2023: Paromita P., Mundnich K., **Booth B.M.**, Narayanan S.S., Chaspari T. "Modeling Inter-individual Differences in Ambulatory-based Multimodal Signals via Metric Learning: A Case study of personalized well-being estimation of healthcare workers." Frontiers in Digital Health, 2023.

2023: **Booth B.**, Bosch N., D'Mello S. "Engagement Detection and its Applications in Learning: A Tutorial and Selective Review." Proceedings of the IEEE, Special Issue on Affective Computing. IEEE, 2023. (IF=14.9)

2022: D'Mello S., **Booth B.** "Affect Detection from Wearables in the 'Real' Wild: Fact, Fantasy, or Somewhere Inbetween?" IEEE Intelligent Systems, 2022. (IF=6.7)

2022: Yau J., Girault B., Feng T., Mundnich K., Nadarajan A., **Booth B.**, Hsieh E., Narayanan S. "TILES-2019, A Longitudinal Physiologic and Behavioral Data Set of Medical Residents in an Intensive Care Unit." Nature Scientific Data, 2022. (IF=6.7)

2022: **Booth B.**, Vrzakova H., Mattingly S., Martinez G., Faust L., D'Mello S. "Toward Robust Stress Prediction in the Age of Wearables: Modeling Perceived Stress in a Longitudinal Study with Information Workers." Transactions on Affective Computing, 2022. (IF=10.7)

2022: Huggins-Manley, C., **Booth B.**, D'Mello S. "Toward Argument-based Fairness with an Application to AI-Enhanced Educational Assessments." Journal of Educational Measurement, 2022. (IF=1.5)

2022: Tay L., Woo S., Hickman L., **Booth B.**, D'Mello S. "A Conceptual Framework for Investigating and Mitigating Machine Learning Bias (MLMB) in Psychological Assessment." Advances in Methods and Practices in Psychological Science, 2022.

2021: **Booth B.**, Hickman L., Subburaj S., Tay L., Woo S., D'Mello S. "Integrating Psychometrics and Computing Perspectives on Bias and Fairness in Affective Computing: A Case Study of Automated Video Interviews." Signal Processing Magazine, Special Issue on Affective Computing. IEEE, 2021. (IF=11.1)

2021: Feng T., **Booth B.**, Baldwin-Rodriguez B., Osorno F., Narayanan S. "A Multimodal Analysis of Physical Activity, Sleep, and Work Shift in Nurses with Wearable Sensor Data." Nature Scientific Reports, 2021. (IF=4.4)

2020: Ravuri V., Paromita P., Mundnich K., Nadarajan A., **Booth B.**, Narayanan S., Chaspari T. "Investigating Group-specific Models of Hospital Workers' Well-being: Implications for Algorithmic Bias." International Journal of Semantic Computing, 2020.

2020: K. Mundnich, **B. M. Booth**, M. L'Hommedieu, J. L'Hommedieu, M. Wildman, T. Feng, B. Girault, S. Skaaden, A. Nadarajan, J. L. Villatte, T. H. Falk, K. Lerman, E. Ferrara, and S. Narayanan, "TILES-2018, A Longitudinal Physiologic and Behavioral Data Set of Hospital Workers", Nature Scientific Data, 2020. (IF=6.7)

2019: Mundnich K., **Booth B.M.**, Girault B., Narayanan S., "Generating Labels for Regression of Subjective Constructs using Triplet Embeddings", Pattern Recognition Letters, 2019. (IF=4.8)

2019: **B. M. Booth**, K. Mundnich, T. Feng, A. Nadarajan, T. H. Falk, J. L. Villatte, E. Ferrara, and S. Narayanan,

"Multimodal Human and Environmental Sensing for Longitudinal Behavioral Studies in Naturalistic Settings: Framework for Sensor Selection, Deployment, and Management", Journal of Medical Internet Research, vol. 21, no. 8, e12832, 2019. (IF=6.0)

Conferences

2026: Yang J., Wang R., **Booth B.M.** "Mobile Gameplay Experience and In-Game Advertising: Examining the Effects of Immersion and Advertising." 2026. (submitted)

2026: Naiyer I., Harrell-Williams L.M., Khulman S.L., Pavlik P., Windsor A., **Booth B.M.** "Uncovering Motivational Trajectories in Digital Learning Environments." Educational Data Mining (EDM), 2026. (submitted)

2025: Farzan F., Naiyer I., Jahan S.A., **Booth B.M.** "Choosing to Learn: Achievement Goals, Immersion, and Competition in Non-Educational Digital Games." Educational Data Mining (EDM), 2025.

2025: Li Y., Kollias D., Chanel G., Fanourakis M., Muszynski M., **Booth B.M.**, Tian L., Perera M., Lai C., Chen H. "HRAI 2025: The 1st Workshop on Holistic and Responsible Affective Intelligence." In Proceedings of the 27th International Conference on Multimodal Interaction (ICMI '25), 2025.

2024: **Booth B.M.**, Jacobs J., Bush J.B., Milne B., Fischhaber T., D'Mello S.K. "Human-tutor Coaching Technology (HTCT): Automated Discourse Analytics in a Coached Tutoring Model." LAK24: 14th International Learning Analytics and Knowledge Conference, 2024. (AR=30%)

2023: Moulder B., **Booth B.**, Abitino A., D'Mello S. "Recurrence Quantification Analysis of Eye Gaze Dynamics During Team Collaboration." LAK23: 13th International Learning Analytics and Knowledge Conference, 2023.

2021: **Booth B.**, Hickman L., Subburaj S., Tay L., Woo S., D'Mello S. "Bias and Fairness in Multimodal Machine Learning: A Case Study of Automated Video Interviews." International Conference on Multimodal Interaction, 2021. (AR=20% for oral)

2020: Ravuri, V., Paromita P., Mundnich K., Nadarajan A., **Booth B.**, Narayanan S., Chaspari T., "Group-specific models of healthcare workers' well-being using iterative participant clustering." 2020 Second International Conference on Transdisciplinary AI (TransAI). IEEE, 2020.

2020: **Brandon M. Booth** and Shrikanth S. Narayanan, "Fifty Shades of Green: Towards a Robust Measure of Inter-annotator Agreement for Continuous Signals", In 2020 International Conference on Multimodal Interaction, 2020. (AR=38%)

2020: **Brandon M. Booth** and Shrikanth S. Narayanan, "Trapezoidal Segment Sequencing: A Novel Approach for Fusion of Human-produced Continuous Annotations", In 2020 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), IEEE, 2020.

2020: G. Hadjiantonis, P. Paromita, K. Mundnich, A. Nadarajan, **B.M. Booth**, S. Narayanan, T. Chaspari, "Dynamical Systems Modeling of Day-to-day Signal-based Patterns of Emotional Self-regulation and Stress Spillover in Highly-demanding Health Professions", In Engineering in Medicine and Biology Society (EMBC), 2020 42nd Annual International Conference of the IEEE, IEEE, 2020.

2020: Tiantian Feng, **Brandon M. Booth**, Shrikanth Narayanan. "Modeling Behavior as Mutual Dependency Between Physiological Signals and Indoor Location in Large-scale Wearable Sensor Study", In 2020 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2020.

2019: **Brandon M. Booth**, Shrikanth S. Narayanan. "Trapezoidal Segmented Regression: A Novel Continuous-scale Real-time Annotation Approximation Algorithm", In Affective Computing Intelligent Interaction (ACII), 2019. (AR=29%)

2019: **Brandon M. Booth**, Tiantian Feng, Abhishek Jangalwa, Shrikanth S. Narayanan. "Toward Robust Interpretable Human Movement Pattern Analysis in a Workplace Setting." IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2019.

2019: Wildman M., Eulogio R., Singh R., Ramirez E., Foschini L., Nadarajan A., **Booth B.**, Mundnich K., Ferrara E., Lerman K., Narayanan S. "Comparison of Heart Rate Measurement Between the Fitbit Charge 2 and OMSignal Smart Garments: A Free Living Study", Society for Behavioral Medicine (SBM), 2019.

2018: **Brandon M. Booth**, Karel Mundnich, and Shrikanth S. Narayanan. "Fusing Annotations with Majority Vote Triplet Embeddings." In Proceedings of the 8th International Workshop on Audio/Visual Emotion Challenge, ACM, 2018.

[winner of gold-standard emotion sub-challenge]

2018: Brandon M. Booth, Taylor J. Seamans, Shrikanth S. Narayanan. "An Evaluation of EEG-based Metrics for Engagement Assessment of Distance Learners." In Engineering in Medicine and Biology Society (EMBC), 2018 40th Annual International Conference of the IEEE, IEEE, 2018.

2018: Tiantian Feng, Amrutha Nadarajan, Colin Vaz, **Brandon Booth**, and Shrikanth Narayanan. "TILES audio recorder: an unobtrusive wearable solution to track audio activity." In Proceedings of the 4th ACM Workshop on Wearable Systems and Applications, pp. 33-38. ACM, 2018. **[Patent pending]**

2018: Brandon M. Booth, Karel Mundnich, and Shrikanth S. Narayanan. "A Novel Method for Human Bias Correction of Continuous-time Annotations." In Acoustics, Speech and Signal Processing (ICASSP), 2018 IEEE International Conference on, IEEE, 2018.

2017: Brandon M. Booth, Asem M. Ali, Shrikanth S. Narayanan, Ian Bennett, and Aly A. Farag. "Toward active and unobtrusive engagement assessment of distance learners." In Affective Computing and Intelligent Interaction (ACII), 2017 Seventh International Conference on, pp. 470-476. IEEE, 2017.

2016: Somandepalli, Krishna, Rahul Gupta, Md Nasir, **Brandon M. Booth**, Sungbok Lee, and Shrikanth S. Narayanan. "Online affect tracking with multimodal kalman filters." In Proceedings of the 6th International Workshop on Audio/Visual Emotion Challenge, pp. 59-66. ACM, 2016.

2016: Brandon M. Booth, Rahul Gupta, Pavlos Papadopoulos, Ruchir Travadi, and Shrikanth S. Narayanan. "Automatic Estimation of Perceived Sincerity from Spoken Language." In INTERSPEECH, pp. 2021-2025. 2016.

Books and Book Chapters

2025: Jacobs J., Suresh A., **Booth B.**, Sumner T., Bush J., Brown C., D'Mello S. "Automating feedback from recorded instructional observations: Using AI to detect and support dialogic teaching." In S. Kelly (Ed), Research Handbook on Classroom Observation. Edward Elgar Publishing, March 2025.

2007: John P. Flynt and **Brandon Booth**. "Unreal Tournament Game Programming for Teens." Boston, MA: Thomson Course Technology, 2007.

Invited Talks

January 2026: Panelist, *Why I Love What I Do and How I Got Here Panel*, Signal Analysis and Interpretation Lab Fest, University of Southern California.

September 2025: Naiyer, I., Booth, B. *Uncovering Motivational Profiles from Digital Tutoring Data*, Institute for Intelligent Systems Lightning Talk, University of Memphis.

August 2025: Panelist, *Trust in AI for Forensic Psychology AI Panel*, American Psychological Association, Denver, Colorado.

May 2025: Booth, B. *Practical Artificial Intelligence Panel Discussion*, INTEGRIT-E Delta Regional Collaboration Conference, University of Memphis, Tennessee.

May 2025: Booth, B. *Rethinking EmotionAI: Ground Trust, Bias, and Social Responsibility in Affective Computing*, FG2025 TrustFAA Conference, Clearwater, Florida.

October 2024: Panelist, *AI Ethics and Education Special Interest Group Panel*. American Education Research Association (virtual event)

June 2024: Windsor, A., **Booth, B.** *A Tutorial and Educational Applications for Generative AI*, West Tennessee STEM Hub for K-12 Educators, University of Memphis.

April 2024: Booth, B. *Machine Intelligence for Modeling Naturalistic Human Behavior and Experiences*, Universitat Autònoma de Barcelona (UAB) Lightning Talk, co-hosted by University of Memphis and UAB.

February 2024: Round table discussion. *Policy Talks 2024: A National Convening on Ethics, Narratives, and Artificial Intelligence*, The Center for Practical Ethics, University of Mississippi.

April 2023: Booth, B. *Introduction to Multimodal Machine Learning*, Introduction to Multimodal Machine Learning Course, Computer & Information Science & Engineering Department, University of Florida.

September 2023: Booth, B. *Machine Intelligence for Modeling Naturalistic Human Behavior and Experiences*, Institute for Intelligent Systems Lightning Talk, University of Memphis.

September 2022: Booth, B. *Machine Intelligence for Naturalistic Human Behavior and Experience Modeling*, Institute of Cognitive Science Colloquium, University of Colorado Boulder.

February 2022: Booth, B. *Integrating Psychometrics and Computing Perspectives on Bias and Fairness in Affective Computing*, Data Justice Lab Seminar Series, Texas A&M Institute of Data Science. (virtual event)

April 2021: Booth, B. *Integrating Psychometrics and Computing Perspectives on Bias and Fairness in Affective Computing: A Case Study of Automated Video Interviews*, AI Institute for Student-AI Teaming, University of Colorado Boulder. (virtual event)

April 2021: Booth, B. *Human Perception and Behavioral Machine Intelligence*, Tech Talk, Applied Invention, LLC. (virtual event)

November 2019: Booth, B. *Why Data Scientists Should Get Their Hands Dirty*, Computer Science Colloquium, Colorado State University.

July 2019: Booth, B. *Identifying and Correcting Human-produced Artifacts in Continuous Annotations of Subjective Constructs*, Tech Talk, Evidation Health, Inc. (virtual event)

Service

Review Services: IEEE Transactions on Affective Computing, IEEE Signal Processing Magazine, IEEE Journal of Selected Topics in Signal Processing, Nature Scientific Data, Nature Mental Health, Journal of Medical Internet Research, Open Journal of Signal Processing, Frontiers in Computer Science, ACM Journal of User Modeling and User-Adapted Interaction, ACM Interactive Mobile Wearable and Ubiquitous Technologies Journal, ACM Transactions on Knowledge Discovery from Data Journal, ACM International Conference on Multimodal Interaction, IEEE International Conference on Acoustics Speech and Signal Processing, IEEE Conference on Affective Computing and Intelligent Interaction, ACM User Modeling Adaptation and Personalization, ACM Multimedia, ACM Computing Surveys, IAPR International Conference on Biometrics, IEEE Engineering in Medicine and Biology Conference, Frontiers in Digital Health

Program Committees: 2024 Association for the Advancement of Artificial Intelligence, 2022-2024 ACM International Conference on Multimodal Interaction, 2022-2024 IEEE International Conference on Affective Computing and Intelligent Interaction (ACII), 2021-2024 ACM User Modeling Adaptation and Personalization (UMAP), 2021 ACII Outstanding Dissertation Award Committee

Teaching and Mentorship

Courses Taught

2023–2026: **Capstone Software Project**, University of Memphis. [undergraduate]

2023–2024: **Introduction to Human-Computer Interaction**, University of Memphis. [undergraduate/graduate]

2022–2023: **Computer Science Colloquium** (teaching assistant), University of Colorado Boulder. [graduate]

2016: **Introduction to Computing** (teaching assistant, recitation lecturer), University of Southern California. [undergraduate]

2015: **Computer Graphics** (teaching assistant and guest lecturer), University of Southern California. [graduate]

2011: **Coding for Games**, (curriculum enhancement and teaching), Art Institute Los Angeles. [undergraduate]

2011: **Game Tools**, (full curriculum development and teaching), Art Institute Los Angeles. [undergraduate]

Outreach

July 2024: CodeCrew Summer Camp Invited Speaker, *University of Memphis*

2021–2022: AI Club Research Mentor and Invited Speaker, *Rock Canyon High School*

2020–2021: Research Mentor, Boulder Valley School District Science Research Seminar, *Boulder Valley High School*.

2020: Robotics Mentor, *Colorado Early College Fort Collins*.

Leadership and Teamwork

2025–2026: Tutorial Co-Chair for the International Conference on Affective Computing and Intelligent Interaction (ACII '26).

2023–2026: Community Web Resources Coordinator for the Association for the Advancement in Affective Computing (AAAC).

2024: Doctoral Symposium Chair for the International Conference on Multimodal Interaction (ICMI '24)

2023: Special Session Co-organizer for the International Conference on Biomedical and Health Informatics (BHI '23): *Trustworthy and responsible data analytics for mental health*.

2020–2022: Project lead for a general purpose machine learning library compatible with Tensorflow, PyTorch, Scikit Learn, and other back-ends, used by *Emotive Computing Lab*.

2018–2022: Co-managed sensor testing, participant testing, installation, deployment, and monitoring for large-scale in situ study of hospital workers (TILES) through the *University of Southern California*. Developed website and hosting services for public data set access.

2014–2015: Managed team of dozens of masters students to develop the multimodal *Experience Lab* for human behavior studies at the *University of Southern California*.

2013: NASA Sample Return Challenge participant, *Worcester Polytechnic Institute*.

- Worked with team of professional roboticists and engineers to develop an autonomous robot capable of locating, collecting, and returning various objects hidden in an unfamiliar environment

2005–2007: Science Olympiad judge for Colorado state-wide competitions

2005–2007: Competitor in the Mathematical Contest in Modeling, *University of Colorado Boulder*

- Modeled traffic flow at toll booths and proposed an optimal toll booth-to-lanes ratio. **Received honorable mention.**
- Explored and constructed measures of fairness for Congressional redistricting to avoid gerrymandering and proposed a fair district assignment. **Received meritorious award.**
- Modeled the spreading of AIDS in Africa and considered the impact of varying degrees of protection and United Nations aid. **Received honorable mention.**